A Recursive Quantum Approximate Optimization Framework for Constrained Combinatorial Optimization: Applications to Maximum Independent Set and Balanced Weighted MaxCut

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Abstract

Constrained combinatorial optimization problems are ubiquitous in science and engineering and remain classically intractable at scale, even for canonical benchmarks such as MaxCut and the Maximum Independent Set (MIS) problem. The Quantum Approximate Optimization Algorithm (QAOA) provides a hybrid variational route to approximate solutions on near-term devices, but fixed-depth QAOA is limited by a persistent trade-off: shallow circuits are tractable but constrained by local light-cone effects, whereas deeper circuits that capture longer-range correlations face increasingly rugged optimization landscapes. This trade-off sharpens under hard constraints, where the two standard strategies penalty terms appended to the cost Hamiltonian, and constraint-preserving mixers restricted to the feasible subspace exchange feasibility guarantees for additional circuit depth. We first derive a closed-form expression for the single-layer ($p = 1$) QAOA cost expectation on an arbitrary MaxCut instance, showing that each edge evolves independently as a function of local vertex degree and shared-triangle count, and we validate this expression numerically against exact simulation. Building on this structural picture, we propose a penalty-based Recursive QAOA (RQAOA) framework that eliminates variables using correlation information extracted from shallow ($p = 1$) circuits while enforcing feasibility during elimination and back-substitution. We instantiate this framework for two constrained problems: MIS, through a modified elimination rule that preserves the independence constraint, and balanced weighted MaxCut, through a weight-merging update suited to graph partitioning under a cardinality-balance constraint. On MIS instances where fixed-depth QAOA fails to converge to the optimum, the proposed RQAOA recovers exact solutions using only $p = 1$ circuits. Applied to a synthetic traffic-network partitioning problem formulated as balanced weighted MaxCut, penalty-based RQAOA attains a perfect approximation ratio of 1.0000 at $p = 1$, exceeding a constraint-preserving XY-mixer QAOA (0.9478 at $p = 3$) and standard penalty QAOA (0.8341 at $p = 5$). These results indicate that recursive, correlation-guided elimination can substitute for circuit depth in constrained combinatorial optimization, offering a practical direction for near-term quantum devices.

Keywords: Quantum Approximate Optimization Algorithm; Recursive QAOA; constrained combinatorial optimization; Maximum Independent Set; weighted MaxCut; NISQ algorithms

1 Introduction

Combinatorial optimization problems (COPs) with constraints underlie tasks ranging from network design to logistics and resource allocation. Most COPs of practical interest are NP-hard: the number of candidate solutions grows as 2^n in the number of decision variables n, and constraints fragment the feasible region further, degrading both exact and heuristic classical methods as n grows [1, 2]. Quantum computing offers an alternative encoding: a COP can be mapped onto the ground state of a problem Hamiltonian, so that ground-state preparation followed by measurement yields a candidate optimal solution. The adiabatic theorem guarantees that a system initialized in the ground state of a simple Hamiltonian remains in the instantaneous ground state under sufficiently slow evolution toward the problem Hamiltonian [3], but the coherence times required for genuinely adiabatic evolution are unavailable on current noisy intermediate-scale quantum (NISQ) hardware.

The Quantum Approximate Optimization Algorithm (QAOA), introduced by Farhi, Goldstone, and Gutmann [4], discretizes this adiabatic path into a shallow, parameterized circuit of depth p, alternating a cost unitary $U_C(\gamma) = e^{-i\gamma H_C}$ with a mixer unitary $U_M(\beta) = e^{-i\beta H_M}$, and delegates parameter selection to a classical outer loop. QAOA is attractive for NISQ hardware because its circuits are shallow and its structure tolerates penalty-based or mixer-based constraint handling [5]. In practice, however, QAOA exhibits a well-documented tension: performance at fixed depth is bounded by the effective light-cone of the ansatz, so instances with large diameter or odd-cycle frustration are poorly approximated at low p [4]; increasing p improves expressibility but simultaneously deepens the parameter landscape, exposing the optimizer to local minima and, at scale, barren plateaus [6, 7].

This tension is compounded when the objective is subject to hard constraints. Two established strategies exist. Penalty-based encoding folds constraint violations into the cost Hamiltonian via a scalar coefficient λ [1, 8]; it is architecture-agnostic and simple to implement, but λ must be tuned empirically, and the resulting energy landscape can become ill conditioned or admit a nontrivial rate of infeasible outcomes. Constraint-preserving (alternating-operator) mixers instead restrict the quantum evolution to the feasible subspace by construction [5], guaranteeing feasibility but at the cost of multi-controlled gates and correspondingly deeper circuits. A third line of work, Recursive QAOA (RQAOA), sidesteps the depth-versus-landscape trade-off altogether by using a shallow circuit only to estimate pairwise correlations $\langle Z_i Z_j \rangle$, then eliminating the most strongly correlated variable pair and repeating on a reduced instance [9]. RQAOA has been shown to outperform fixed-depth QAOA on unconstrained MaxCut, including on complete graphs [10], but its extension to problems with genuine hard constraints where elimination and back-substitution must themselves preserve feasibility is comparatively underexplored.

This paper addresses that gap. We make four contributions:

1. We derive a closed-form expression for the exact $p = 1$ QAOA cost expectation value on an arbitrary MaxCut instance, expressed in terms of local vertex degree and shared-triangle count, and confirm it numerically against statevector simulation on a range of small graphs and special graph families.

2. We characterize, quantitatively, the trade-off between penalty-based and constraint-preserving strategies for a canonical hard-constraint problem, the Maximum Independent Set (MIS), including their respective approximation ratios, feasibility rates, and circuit-depth requirements.

3. We propose a penalty-based RQAOA framework in which the elimination and back-substitution rules are made constraint-aware: a modified elimination rule for MIS that avoids assignments violating the independence constraint, and a weight-merging elimination rule for balanced weighted

MaxCut that preserves both the cut objective and a cardinality-balance constraint under vertex removal.

4. We apply the proposed framework to a synthetic traffic-network partitioning problem, formulated as balanced weighted MaxCut on a grid graph, and provide a direct, matched-instance comparison of penalty QAOA, constraint-preserving mixer QAOA, and penalty-based RQAOA in terms of approximation ratio and circuit depth.

The remainder of the paper is organized as follows. Section 2 surveys classical and quantum approaches to constrained optimization and positions RQAOA relative to prior constraint-handling strategies. Section 3 fixes notation and reviews the QUBO-to-Ising mapping, the QAOA ansatz, and the standard RQAOA procedure, and presents the closed-form single-layer expectation value used throughout. Section 4 presents the proposed penalty-based RQAOA framework, its constraint encodings for MIS and balanced weighted MaxCut, and the unified algorithm. Section 5 details the simulation environment and problem instances. Section 6 reports and discusses the results. Section 7 states the scope and limitations of the present study, and Section 8 concludes.

2 Background and Related Work

Classical strategies for constrained combinatorial optimization fall into two families. Exact methods branch-and-bound and dynamic programming guarantee optimality but scale exponentially in the worst case [11]. Heuristic and metaheuristic methods, including simulated annealing, genetic algorithms, and penalty-based local search, scale more favorably but offer no optimality guarantee and typically require problem-specific tuning [11, 2]. Both families degrade as constraints fragment the feasible region into an exponentially small subset of the full search space [2, 12].

Quantum adiabatic computation reframes optimization as ground-state preparation under a Hamiltonian that interpolates slowly from an easily prepared initial Hamiltonian to a problem Hamiltonian, with correctness guaranteed by the adiabatic theorem [13, 3]. Direct implementation demands evolution times and coherence far beyond current NISQ hardware, motivating variational, Trotterized approximations [14, 15]. QAOA [4] instantiates this idea as a shallow, parameterized circuit optimized in a hybrid quantum-classical loop and has become a default variational approach for combinatorial optimization on NISQ devices; Blekos et al. survey a broad family of extensions, including multi-angle QAOA, digitized counterdiabatic driving [16], and classically warm-started initialization [17, 18].

For constrained problems specifically, two constraint-handling paradigms dominate the literature. Penalty methods absorb constraint violations into the cost Hamiltonian through a scalar coefficient, following the general QUBO-penalty methodology of Lucas [1] and Glover et al. [8]; this preserves the standard transverse-field mixer and circuit structure, but leaves the penalty weight as an open, instance-dependent tuning parameter that trades feasibility against landscape conditioning. Constraint-preserving mixers, formalized as the Quantum Alternating Operator Ansatz by Hadfield et al. [5, 19], instead design a mixer Hamiltonian that commutes with the constraint operator and confines the evolution to the feasible subspace by construction, guaranteeing feasible measurement outcomes at the cost of additional multi-controlled gates.

A separate line of work reduces problem size rather than modifying the ansatz. Bravyi et al. [9] introduced Recursive QAOA (RQAOA), in which a shallow QAOA circuit is used only to estimate the sign and magnitude of pairwise correlations $\langle Z_i Z_j \rangle$; the most strongly correlated pair is then fixed and eliminated, and the process repeats on the resulting smaller instance. This decouples solution quality from ansatz depth and has been shown to outperform fixed-depth QAOA on unconstrained

MaxCut, including on complete graphs where standard QAOA is known to perform poorly [10]. Existing treatments of RQAOA, however, are formulated for unconstrained or already-penalized objectives: elimination and back-substitution assign variable values based solely on correlation sign, with no explicit mechanism for preserving a hard constraint such as independence or a target partition size during the reduction step itself. It is not automatic that a correlation-driven elimination rule designed for an unconstrained cut objective remains feasibility-preserving once independence or balance constraints are introduced. The present work addresses this gap directly: we make the elimination and back-substitution rules constraint-aware for two representative constrained problems, and evaluate the resulting framework against both established constraint-handling baselines and a practical partitioning application.

3 Theoretical Foundations and Analytical Analysis

3.1 QUBO Formulation

A Quadratic Unconstrained Binary Optimization (QUBO) problem is specified by a cost function over binary decision variables $x \in \{0,1\}^n$,

$$C(x) = \sum_{i,j=1}^{n} Q_{ij}\, x_i x_j + \sum_{i=1}^{n} c_i x_i, \tag{1}$$

where $Q \in \mathbb{R}^{n\times n}$ and $c \in \mathbb{R}^n$. Since $x^\top Q x = x^\top \left(\frac{Q+Q^\top}{2}\right) x$, Q may be taken symmetric without loss of generality. QUBO is the standard intermediate representation for mapping combinatorial problems including those with linear equality or inequality constraints, absorbed via penalty terms onto quantum hardware [1, 8].

3.2 Ising Hamiltonian

To express Eq. (1) in the language of a quantum Hamiltonian, we apply the affine change of variables from binary to spin variables,

$$s_i = 2x_i - 1 \in \{-1,+1\} \iff x_i = \frac{s_i+1}{2}. \tag{2}$$

Substituting Eq. (2) into Eq. (1) and discarding the resulting spin-independent constant (the Hamiltonian is defined up to an additive shift) yields

$$H(s) = \sum_{\substack{i,j=1 \\ i\le j}}^{n} a_{ij}\, s_i s_j + \sum_{i=1}^{n} b_i\, s_i, \tag{3}$$

with coefficients

$$a_{ij} = \begin{cases} Q_{ij}/2, & i \neq j \\ Q_{ii}/4, & i = j \end{cases}, \qquad b_i = c_i + \frac{1}{2}\sum_{j=1}^{n} Q_{ij}. \tag{4}$$

Promoting each spin to a Pauli-Z operator on qubit i, $s_i \mapsto Z_i$, gives the cost (problem) Hamiltonian

$$H_C = \sum_{\substack{i,j=1 \\ i\le j}}^{n} a_{ij}\, Z_i Z_j + \sum_{i=1}^{n} b_i\, Z_i, \tag{5}$$

which is diagonal in the computational basis, with eigenvalues equal to the classical QUBO cost of the corresponding bitstring.

3.3 Quantum Approximate Optimization Algorithm

3.3.1 Ansatz and Hybrid Optimization Loop

QAOA prepares a variational state by alternating p layers of the cost unitary $U_C(\gamma) = e^{-i\gamma H_C}$ with a mixer unitary $U_M(\beta) = e^{-i\beta H_M}$, where the standard mixer is the transverse field $H_M = \sum_{i=1}^{n} X_i$. Starting from the uniform superposition $|+\rangle^{\otimes n} = H^{\otimes n}|0\rangle^{\otimes n}$ the unique maximal eigenstate of H_M, and hence a natural, trivially prepared reference state the depth-p ansatz is

$$|\gamma, \beta\rangle = \prod_{l=1}^{p} U_M(\beta_l)\, U_C(\gamma_l)\, |+\rangle^{\otimes n}, \qquad \gamma, \beta \in \mathbb{R}^p, \tag{6}$$

and a classical optimizer adjusts (γ, β) to minimize $\langle H_C\rangle = \langle \gamma, \beta | H_C | \gamma, \beta\rangle$. This construction can be understood as a first-order Trotterization of the adiabatic path between H_M and H_C, with the fixed adiabatic schedule replaced by free variational parameters at each layer, which allows the optimizer to depart from the adiabatic trajectory and reach comparable solution quality with far shorter evolution [4].

At the gate level, $U_M(\beta)$ factorizes into single-qubit rotations, $U_M(\beta) = \bigotimes_{i=1}^{n} R_x^{(i)}(2\beta)$, because the terms of H_M commute. $U_C(\gamma)$ is diagonal in the computational basis and factorizes into commuting single-and two-qubit phase gates: each linear term $b_i Z_i$ contributes a single-qubit R_z rotation, and each quadratic term $a_{ij} Z_i Z_j$ contributes a two-qubit R_{ZZ} interaction, implementable as $\mathrm{CNOT}(i,j)\,[I \otimes R_z(2\gamma a_{ij})]\,\mathrm{CNOT}(i,j)$ [20]. The complete hybrid procedure is summarized in Algorithm 1.

Algorithm 1 Quantum Approximate Optimization Algorithm

Require: Cost Hamiltonian H_C, depth p, shot count N_s, initial parameters $\gamma^{(0)}, \beta^{(0)}$
Ensure: Optimized parameters γ^*, β^* and an approximate solution bitstring
1: Prepare $|+\rangle^{\otimes n}$ via Hadamard gates on all qubits
2: **repeat**
3: Construct the depth-p circuit of Eq. (6) with current parameters (γ, β)
4: Measure all qubits N_s times, obtaining bitstrings $z^{(1)}, \ldots, z^{(N_s)}$
5: Compute the empirical cost $\hat{F}_p(\gamma, \beta) = \frac{1}{N_s}\sum_{k=1}^{N_s} C\big(z^{(k)}\big)$
6: Update $(\gamma, \beta) \leftarrow \textsc{ClassicalOptimizer}\Big(\hat{F}_p, \gamma, \beta\Big)$
7: **until** convergence or a maximum iteration budget is reached
8: Run the final circuit at the optimized parameters (γ^*, β^*)
9: **return** the most frequently measured bitstring as the approximate solution

3.3.2 Exact Single-Layer Expectation Value for MaxCut

For the MaxCut problem on a graph $G(V, E)$ with $|V| = n$, $|E| = m$, the cost Hamiltonian is $H_C = \sum_{(i,j)\in E} \frac{1}{2}(I - Z_i Z_j)$, and the approximation ratio of a state $|\psi\rangle$ is $r = \langle\psi|H_C|\psi\rangle/\mathrm{MaxCut}(G)$. Because H_C is a sum over edges, $\langle H_C\rangle = \sum_{(i,j)\in E} \frac{1}{2}\,(1 - \langle Z_i Z_j\rangle)$, so the expectation value is fully determined by the pairwise correlators $\langle Z_i Z_j\rangle$. At depth $p = 1$ these correlators admit an exact closed form.

Proposition 1 (Exact $p = 1$ MaxCut expectation value)**.** *Let $G(V, E)$ be an arbitrary simple graph, and consider the depth-one QAOA state $|\gamma, \beta\rangle = U_M(\beta)U_C(\gamma)|+\rangle^{\otimes n}$ with $H_C = \sum_{(i,j)\in E} \frac{1}{2}(I - Z_i Z_j)$. For an edge $(i, j) \in E$, write d_i, d_j for the degrees of its endpoints and $f = |N(i) \cap N(j)|$ for* *the number of common neighbors (triangles containing the edge). Then, up to the local reparameterization $U_C(\gamma) \propto \bigotimes_{(a,b)\in E}(\cos\gamma + i\sin\gamma\, Z_a Z_b)$ used below,*

$$\langle H_C\rangle = \frac{m}{2} + \frac{1}{4}\sin 4\beta\,\sin\gamma \sum_{(i,j)\in E}\Big(\cos^{d_i}\gamma + \cos^{d_j}\gamma\Big) - \frac{1}{4}\sin^2 2\beta \sum_{(i,j)\in E}\cos^{d_i+d_j-2f}\gamma\,\Big(1-\cos^f 2\gamma\Big). \quad (7)$$

Derivation sketch. Conjugating Z_iZ_j through the mixer gives

$$e^{i\beta H_M} Z_i Z_j\, e^{-i\beta H_M} = (\cos 2\beta\, Z_i + \sin 2\beta\, Y_i)\,(\cos 2\beta\, Z_j + \sin 2\beta\, Y_j)\,; \quad (8)$$

since $U_C(\gamma)|+\rangle^{\otimes n}$ has definite parity under each Z-basis flip and U_C is diagonal, only the Z_iZ_j term of Eq. (8) survives expectation, giving $\langle Z_iZ_j\rangle = \cos^2 2\beta\,\langle\phi|Z_iZ_j|\phi\rangle$ with $|\phi\rangle = U_C(\gamma)|+\rangle^{\otimes n}$. Expanding $U_C(\gamma)$ as a product over edges and evaluating $\langle\phi|Z_iZ_j|\phi\rangle$ as a sum over edge subsets $S \subseteq E$, only subsets S for which every vertex has even incidence except i and j, which are forced to odd incidence by the fixed pair contribute; to leading order this restricts S to the direct edge (i,j) together with even-sized subsets of the f two-edge paths through common neighbors. Summing this restricted combinatorial series in closed form, and reintroducing the mixer-angle dependence, yields Eq. (7).

Equation (7) shows that, at $p = 1$, each edge's contribution to $\langle H_C\rangle$ depends only on the local subgraph around that edge its endpoint degrees and the number of triangles it participates in and not on the graph's global structure. This is the analytic content of the well-known light-cone argument for constant-depth QAOA [4, 21]: at $p = 1$ the effective causal neighborhood of an edge extends only to its immediate neighbors. Table 1 lists the resulting approximation ratio for several structurally distinct graph families obtained by optimizing Eq. (7) over (γ,β).

Table 1: Optimal $p = 1$ QAOA approximation ratio for MaxCut on structurally distinct graph families, obtained from Eq. (7). D denotes vertex degree for large-girth D-regular graphs, and n the cycle length for C_n.

Graph family	MaxCut value	Approximation ratio r
Bipartite graphs	m	1.000
Triangle (K_3 / C_3)	2	0.750
Even cycles (C_{2k})	n	1.000
Odd cycles (C_{2k+1})	$n-1$	$4(n-1)/3n$
D-regular, large girth		$\geq \frac{1}{2} + \frac{1}{2\sqrt{D}}\left(\frac{D-1}{D}\right)^{(D-1)/2}$
Complete graphs K_n ($n\to\infty$)	$\approx n^2/4$	$\to 0.500$

The $D = 3$ case bounds the approximation ratio at $r \geq 0.6924$, recovering the value originally reported for $p = 1$ QAOA on large-girth 3-regular graphs [4]; the triangle case gives the exact optimum $r = 0.75$ at $(\gamma^*,\beta^*) = (\pi/4,\pi/4)$. As a complementary, fully independent check, the minimal two-qubit instance (G a single edge) can be solved exactly by direct diagonalization: with $H_C = \frac{1}{2}(I - Z_1Z_2)$, the exact expectation value is $\langle H_C\rangle(\gamma,\beta) = \frac{1}{2}(1+\sin 2\beta\sin\gamma)$, maximized at $(\gamma^*,\beta^*) = (\pi/2,\pi/4)$ with $\langle H_C\rangle_{\max} = 1$, i.e. a single QAOA layer solves the two-vertex instance exactly. Section 6 validates Eq. (7) numerically against statevector simulation.

3.4 Standard Recursive QAOA (RQAOA)

Equation (7) makes explicit why fixed-depth QAOA at $p = 1$ is structurally limited: correlations are local, so instances whose optimal solution requires coordinating variables beyond the immediate neighborhood long or odd cycles, for instance cannot be resolved without increasing p, which in turn deepens the optimization landscape. Recursive QAOA (RQAOA) [9] avoids this trade-off by using only shallow circuits, but recovering long-range structure through repeated correlation estimation and variable elimination rather than through circuit depth. Algorithm 2 states the standard procedure for MaxCut.

Algorithm 2 Recursive QAOA (RQAOA) for MaxCut

Require: Graph $G(V, E)$, QAOA depth p, correlation threshold τ
Ensure: Binary assignment $x \in \{0, 1\}^{|V|}$

while $|V| > 2$ **do**
  Run QAOA at depth p on the current graph and obtain optimized parameters (γ^*, β^*)
  Compute $M_{ij} = \langle Z_i Z_j \rangle$ at (γ^*, β^*) for every edge $(i, j) \in E$
  $(i^*, j^*) \leftarrow \arg\max_{(i,j)\in E} |M_{ij}|$
  if $|M_{i^*j^*}| < \tau$ **then**
    break
  end if
  Impose $x_{j^*} = x_{i^*}$ if $M_{i^*j^*} > 0$, else $x_{j^*} = 1 - x_{i^*}$
  Remove vertex j^* and its incident edges from G
end while
Solve the remaining ($\leq$ 2-vertex) instance exactly by brute force
Back-substitute eliminated variables using the recorded relations
return x

Because each recursive step requires only a $p = 1$ (or otherwise fixed, shallow) QAOA evaluation, RQAOA trades $O(|V|)$ sequential shallow-circuit evaluations for the deep, highly non-convex landscape that a single large-p QAOA circuit would otherwise require. As reviewed in Section 2, RQAOA in this form assumes an unconstrained (or already fully penalized) objective, with no explicit mechanism for enforcing a hard constraint during elimination. Constraint handling in QAOA more broadly follows one of two strategies penalty-based encoding, which appends $\lambda H_{\text{penalty}}$ to H_C at the cost of a tunable coefficient λ, or constraint-preserving mixers, which replace H_M with an operator that commutes with the constraint [5]. Section 4 develops a version of RQAOA in which the elimination and back-substitution rules are made explicitly compatible with both constraint types considered in this work.

4 Constraint Aware Recursive QAOA Framework

4.1 Problem Formulation

We consider constrained combinatorial optimization problems of the general form

$$\max_{x\in\{0,1\}^n} f(x) \quad \text{subject to} \quad x \in \mathcal{F} \subseteq \{0,1\}^n, \tag{9}$$

where f is a graph-defined objective and $\mathcal{F}$ is a feasible set defined by one or more constraints. We instantiate Eq. (9) for two representative problems that differ in constraint type a hard, structural constraint and a soft, cardinality constraint to test the generality of the proposed framework.

Maximum Independent Set (MIS). Given $G(V, E)$, an independent set is $S \subseteq V$ such that no two vertices of S are adjacent. With $x_i = 1$ indicating $i \in S$,

$$\max_{x \in \{0,1\}^n} \sum_{i=1}^{n} x_i \quad \text{subject to} \quad x_i x_j = 0 \;\; \forall (i,j) \in E. \tag{10}$$

The constraint $x_i x_j = 0$ is a *hard* constraint: any assignment that violates it is infeasible regardless of objective value, so the encoding must either penalize or structurally forbid violation.

Balanced weighted MaxCut. Given a weighted graph $G(V, E, w)$ with $w_{ij} \geq 0$ representing, in our application, road-segment traffic intensity, and $x_i \in \{0, 1\}$ indicating the phase assignment of intersection i,

$$\max_{x \in \{0,1\}^n} \sum_{(i,j) \in E} w_{ij} \left(x_i + x_j - 2x_i x_j\right) \quad \text{subject to} \quad \left| \sum_{i=1}^{n} x_i - \frac{n}{2} \right| \leq \delta, \tag{11}$$

for a small integer tolerance δ. Unlike the MIS constraint, this is a *soft, aggregate* constraint on the cardinality of the partition rather than a pairwise structural one, so it is naturally suited to a squared-penalty encoding.

Both problems share a common template: an unconstrained cut- or cardinality-type objective, plus a feasibility requirement that any solution method including a recursive elimination procedure must respect. Section 4.2 formalizes the two constraints as penalty terms, Section 4.3 converts the resulting QUBOs to Ising form, and Section 4.4 shows how the elimination and back-substitution rules of Algorithm 2 must be modified for each constraint type to remain feasibility-preserving.

4.2 Penalty-Based Constraint Encodings

Independence constraint (MIS). We adopt a penalty-based encoding: violations of $x_i x_j = 0$ are penalized by adding $\lambda \sum_{(i,j) \in E} x_i x_j$ to the (negated, minimization-form) objective, with $\lambda > 0$ chosen larger than the maximum possible gain from violating a single constraint ($\lambda > 1$ suffices, since each edge violation can increase the unpenalized objective by at most 1); we use $\lambda = 2$ throughout. As reviewed in Section 3.4, an alternative is a constraint-preserving mixer that confines the evolution to the independent-set subspace by construction; Section 6 compares both strategies directly before motivating the recursive extension.

Balance constraint (weighted MaxCut). We encode the aggregate cardinality constraint of Eq. (11) as a squared-deviation penalty,

$$\Big(\sum_{i=1}^{n} x_i - \tfrac{n}{2} \Big)^2 = \sum_{i=1}^{n} x_i + 2 \sum_{i<j} x_i x_j - n \sum_{i=1}^{n} x_i + \frac{n^2}{4} = 2 \sum_{i<j} x_i x_j - (n-1) \sum_{i=1}^{n} x_i + \frac{n^2}{4}, \tag{12}$$

using $x_i^2 = x_i$ for binary variables. This is a dense, all-pairs quadratic term distinct from the sparse, edge-local independence penalty used for MIS and couples every pair of qubits, not only those adjacent in G.

4.3 Derivation of Problem-Specific Penalty Hamiltonians

MIS. The penalized, minimization-form QUBO is $C(x) = -\sum_i x_i + \lambda \sum_{(i,j) \in E} x_i x_j$. Substituting $x_i = (1 - Z_i)/2$ and discarding additive constants gives the Ising cost Hamiltonian

$$H_C^{\mathrm{MIS}} = \sum_{i=1}^{n} \left(\frac{1}{2} - \frac{\lambda d_i}{4} \right) Z_i \; + \; \frac{\lambda}{4} \sum_{(i,j) \in E} Z_i Z_j, \tag{13}$$

where d_i is the (unweighted) degree of vertex i; the standard transverse-field mixer $H_M = \sum_i X_i$ is retained.

Balanced weighted MaxCut. Combining the negated cut objective with λ_B times Eq. (12), the total minimization-form QUBO is

$$C_{\text{total}}(x) = -\sum_{i=1}^{n} d_i^w x_i + \sum_{i<j} \left[2w_{ij}\, I_{(i,j)\in E} + 2\lambda_B\right] x_i x_j \;+\; \lambda_B \left[-(n-1)\sum_{i=1}^{n} x_i\right] + \text{const}, \tag{14}$$

where $d_i^w = \sum_{j:(i,j)\in E} w_{ij}$ is the weighted degree of vertex i. Applying $x_i = (1 - Z_i)/2$ to Eq. (14), the linear-in-Z_i contributions from the cut term and the balance term cancel exactly (each vertex's weighted degree contribution is offset by its share of the all-pairs balance penalty), leaving a purely quadratic Ising Hamiltonian,

$$H_C^{\text{bal}} = \sum_{(i,j)\in E} \frac{w_{ij}}{2}\, Z_i Z_j \;+\; \frac{\lambda_B}{2} \sum_{i<j} Z_i Z_j \;+\; \text{const}. \tag{15}$$

Because the second sum in Eq. (15) runs over *all* vertex pairs rather than only edges of G, it is implemented by treating the graph as fully connected for the purpose of the balance term: pairs already in E receive an effective coupling $w_{ij}/2 + \lambda_B/2$, and all remaining pairs receive $\lambda_B/2$. This Hamiltonian, together with the standard mixer, defines the penalty-based QAOA circuit used for balanced weighted MaxCut throughout this work; Section 6 additionally reports a constraint-preserving variant that instead restricts the mixer to the fixed-Hamming-weight (balanced) subspace via an XY-mixer, $H_M^{XY} = \sum_{i<j} (X_i X_j + Y_i Y_j)$, which commutes with the total magnetization $\sum_i Z_i$ and therefore preserves partition size exactly, at the cost of two-qubit XY interactions in place of the single-qubit transverse field.

4.4 Feasibility-Preserving Elimination and Substitution Rules

The standard RQAOA elimination rule (Algorithm 2) assigns $x_{j^*} = x_{i^*}$ or $x_{j^*} = 1 - x_{i^*}$ based only on the sign of the strongest correlation, which is sufficient for an unconstrained objective but not, in general, feasibility-preserving once a hard or aggregate constraint is present. We modify the elimination and back-substitution rules separately for each target problem.

Modified elimination rule for MIS. In the independent-set subspace, a correlation $M_{ij} \to -1$ indicates that i and j tend to take opposite values exactly one of them is selected which is always compatible with independence. A correlation $M_{ij} \to +1$ indicates that i and j tend to agree; since both being selected would violate the constraint whenever $(i, j) \in E$, naively setting $x_j = x_i$ on strong positive correlation risks propagating an infeasible assignment. We therefore restrict elimination to strong negative correlations,

$$M_{i^*j^*} < -\tau \implies x_{j^*} = 1 - x_{i^*}, \quad \text{eliminate } j^*; \tag{16}$$

if the strongest correlation is positive, that edge is skipped in favor of the next-strongest negative correlation, and the recursion terminates (proceeding to the brute-force base case) once no edge exceeds the threshold in magnitude with negative sign. This guarantees that every recursive step is independence-preserving by construction, at the cost of leaving some strongly-but-positively-correlated pairs unresolved by elimination.

Weight-merging elimination rule for balanced weighted MaxCut. Here both signs of correlation are informative and usable $M_{ij} > 0$ indicates i, j tend to share a phase, $M_{ij} < 0$ indicates opposite phases so elimination proceeds on the strongest $|M_{ij}|$ regardless of sign, as in standard

RQAOA. However, removing vertex j^* must preserve both the cut objective and the running balance target on the reduced graph. We therefore: (i) set $x_{j^*} = x_{i^*}$ if $M_{i^*j^*} > 0$ and $x_{j^*} = 1 - x_{i^*}$ otherwise; (ii) for every remaining neighbor k of j^*, merge its incident weight into the corresponding edge at i^*,

$$w'_{i^*k} \; = \; w_{i^*k} \; + \; w_{j^*k} \cdot \operatorname{sign}(M_{i^*j^*}), \tag{17}$$

introducing the edge (i^*, k) if it did not already exist, so that the reduced graph's optimal cut value, once x_{j^*} is back-substituted, equals that of the original graph; and (iii) decrement the running balance target by the assigned value of x_{j^*}, so the remaining variables are recursively balanced against an updated (integer) target rather than the original $n/2$. This rule generalizes the unconstrained RQAOA elimination step (Section 3.4) with an explicit weight-update and a constraint-propagation step, rather than replacing it.

4.5 Unified Constraint-Aware RQAOA Algorithm

Algorithm 3 unifies both instantiations into a single constraint-aware, penalty-based RQAOA procedure, parameterized by a problem type $\Pi \in \{\text{MIS}, \text{Balanced-MaxCut}\}$ that selects the corresponding penalty Hamiltonian (Eqs. (13) or (15)) and elimination rule (Eq. (16) or Eqs. (17)).

Algorithm 3 Constraint-Aware Penalty-Based Recursive QAOA

Require: Graph $G(V, E[, w])$, problem type Π, QAOA depth p, penalty coefficient(s), threshold τ
Ensure: Feasible assignment $x \in \{0, 1\}^{|V|}$

```
Build the penalty Hamiltonian H_C(G) for problem type Π                ▷ Eq. (13) or (15)
while |V| exceeds the brute-force base-case size do
    Run QAOA at depth p on H_C(G) with mixer H_M = ∑_i X_i; obtain (γ*, β*)
    Compute M_ij = ⟨Z_i Z_j⟩ at (γ*, β*) for every edge (i, j) of the current graph
    (i*, j*) ← arg max_(i,j) |M_ij| subject to the sign restriction of problem type Π
    if no eligible edge has |M_i*j*| ≥ τ then
        break
    end if
    if Π = MIS then
        Set x_j* = 1 − x_i*                ▷ Eq. (16); positive-correlation edges are skipped
    else[Π = Balanced-MaxCut]
        Set x_j* = x_i* if M_i*j* > 0, else x_j* = 1 − x_i*
        Update neighbor weights w'_i*k ← w_i*k + w_j*k · sign(M_i*j*) for all k ∈ N(j*) \ {i*}    ▷ Eq. (17)
        Decrement the running balance target by x_j*
    end if
    Remove vertex j* and its incident edges; rebuild H_C(G) on the reduced graph
end while
Solve the remaining base-case instance exactly by brute force, respecting the constraint of type Π
Back-substitute all eliminated variables using the recorded relations
return x
```

By construction, every elimination step of Algorithm 3 preserves feasibility with respect to the target constraint, while retaining the central property of RQAOA that only shallow ($p = 1$ in all experiments reported here) QAOA circuits are evaluated at any point in the recursion.

5 Numerical Simulation and Experimental Setup

5.1 Quantum Simulation Environment and Solvers

All circuits are simulated with Qiskit Aer: expectation values and correlations are computed exactly using the noiseless `StatevectorSimulator`, and sampling-based feasibility rates use the `QasmSimulator` with 8192 shots. Variational parameters are optimized per layer over $\gamma_l \in [0, \pi]$, $\beta_l \in [0, \pi/2]$ using a coarse 20×20 grid search followed by local refinement with COBYLA (convergence tolerance 10^{-6}, 10 random restarts to mitigate local minima). Problem sizes are restricted to $n \leq 8$ vertices, consistent with exact statevector simulation on classical hardware; all reported results are noiseless, isolating algorithmic from device-level performance.

5.2 Maximum Independent Set

We evaluate penalty-based QAOA, the constraint-preserving mixer, and constraint-aware RQAOA on small MIS instances with known ground truth: a 3-vertex path P_3 (MIS size 2), a 4-cycle C_4 (MIS size 2), and a triangle K_3 (MIS size 1), at depths $p = 1, 2$, with penalty coefficient $\lambda = 2$. RQAOA is additionally evaluated on a 4-vertex path P_4 (MIS size 2) and a 5-cycle C_5 (MIS size 2) against standard penalty QAOA at $p = 2$, using inner depth $p = 1$ throughout.

5.3 Balanced Weighted MaxCut

The traffic-partitioning application is modeled as a synthetic 2×3 planar grid graph ($n = 6$ vertices, $m = 7$ edges, Fig. 1), with intersections as vertices, road segments as edges, and synthetic scalar edge weights encoding relative traffic intensity. Because n is even, an exactly balanced bipartition has 3 vertices per side; we relax this to a tolerance of one, $|\sum_i x_i - 3| \leq 1$, admitting partition sizes $(2, 4)$ or $(3, 3)$. For the penalty-based methods (standard QAOA and RQAOA), the balance term of Eq. (15) is included with $\lambda_B = 2$. For the constraint-preserving method, an XY-mixer confines the 6-qubit evolution to the subspace $\mathcal{S}_c = \{x \in \{0, 1\}^6 : 2 \leq \sum_i x_i \leq 4\}$, with the circuit initialized in a uniform superposition over $\mathcal{S}_c$ rather than $|+\rangle^{\otimes 6}$. All three methods penalty QAOA, constraint-preserving mixer QAOA, and penalty-based RQAOA are evaluated at depths $p = 1$ through 5 on the identical graph instance.

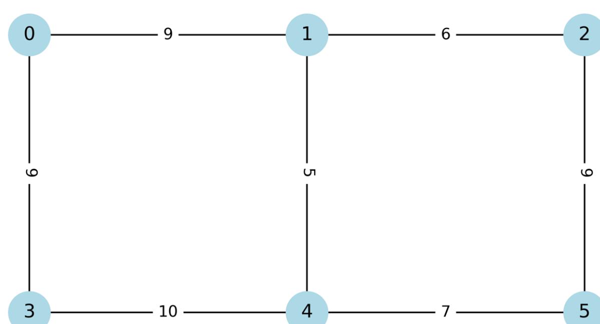


Figure 1: Synthetic traffic routing network used for the balanced weighted MaxCut experiments: a 2×3 planar grid with $n = 6$ vertices and $m = 7$ edges. Edge weights encode relative traffic intensity and define the problem instance used for all three solution methods of Section 5.3.

5.4 Performance Metrics and Baselines

For MaxCut-type objectives we report the approximation ratio $r = \langle \text{objective} \rangle / \text{OPT}$, where OPT is obtained by brute-force enumeration on these small instances. For MIS, we report r as the largest feasible independent-set size found, normalized by the true MIS size; feasibility is enforced by post-selection under the penalty method (discarding measurement outcomes with adjacent selected vertices) and is automatic under the constraint-preserving mixer and under RQAOA, whose elimination rule is feasibility-preserving by construction. We additionally report the *feasibility rate* the fraction of measurement outcomes satisfying the relevant constraint and, where informative, circuit depth in CNOT count, as a proxy for near-term hardware cost.

6 Results and Discussion

We first validate the two structural components underlying the proposed framework the closed-form single-layer expectation value of Proposition 1 and the RQAOA elimination mechanism of Algorithm 2 on unconstrained MaxCut, before evaluating the full constraint-aware framework on MIS and balanced weighted MaxCut.

Table 2 compares the analytical prediction of Eq. (7) against exact statevector simulation at $p = 1$ for six small graphs. Agreement is to within numerical (floating-point) precision in every case, confirming the derivation of Section 3.3.2. Table 3 then compares standard QAOA against RQAOA (inner depth $p = 1$) on four structurally distinct unweighted graphs; on a further five-vertex test graph containing both a 4-cycle and a triangle substructure, RQAOA attains the exact optimum ($r = 1.0000$, single layer) while standard QAOA plateaus at $r = 0.9347$ even at $p = 5$, and on a four-vertex cycle RQAOA reaches the optimal cut of 4 against QAOA's 3 (Table 4), at roughly 3.7× the wall-clock cost of a single QAOA evaluation. These baseline results reproduce the depth-versus-recursion trade-off that motivates the constrained extension developed in Section 4.

Table 2: Analytical ($p = 1$) versus simulated ($p = 1$) cost-Hamiltonian expectation value $\langle H_C \rangle$ for MaxCut on small graphs, validating Proposition 1. All simulated values match the analytical prediction to within 10^{-10}.

Graph	m	MaxCut	Analytical $\langle H_C \rangle$	Simulated $\langle H_C \rangle$	r
Two-vertex edge	1	1	1.0	1.0	1.000
Path P_3	2	2	2.0	2.0	1.000
Triangle K_3	3	2	1.5	1.5	0.750
Cycle C_4	4	4	4.0	4.0	1.000
Square + diagonal	5	4	2.5	2.5	0.625
Complete graph K_4	6	4	3.0	3.0	0.750

6.1 Maximum Independent Set

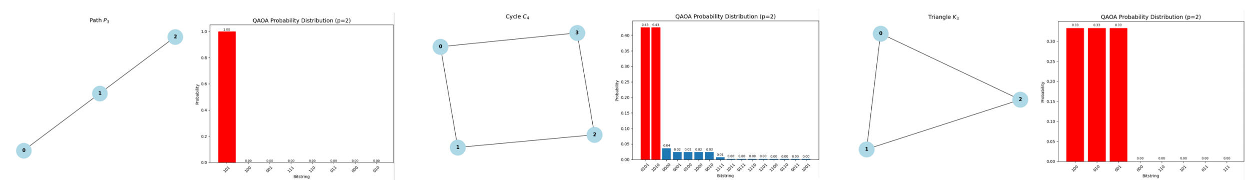
Table 5 reports penalty-based QAOA results. The penalty method recovers the exact MIS on P_3 and K_3 already at $p = 1$, but on C_4 reaches only half the optimal set size at $p = 1$ ($r = 0.5$), improving to $r = 0.75$ at $p = 2$; feasibility rates (the fraction of measurement outcomes with no adjacent selected vertices) are modest throughout (0.550.72), so a nontrivial fraction of measurement shots must be discarded by post-selection.

Table 3: Approximation ratio for standard QAOA ($p = 1, 2$) versus RQAOA (inner depth $p = 1$) on unconstrained MaxCut, unweighted graphs.

Graph	n	QAOA ($p = 1$)	QAOA ($p = 2$)	RQAOA (inner $p = 1$)
C_4	4	0.750	1.000	1.000
C_5	5	0.625	0.750	0.875
C_6	6	0.667	0.778	0.833
K_4	4	0.750	0.920	0.920

Table 4: QAOA versus RQAOA on a four-vertex cycle graph C_4: achieved cut value, optimal cut, approximation ratio, and wall-clock optimization time.

Algorithm	Cut value	Optimal cut	Approximation ratio	Time (s)
QAOA	3	4	0.7500	0.003030
RQAOA	4	4	1.0000	0.011361



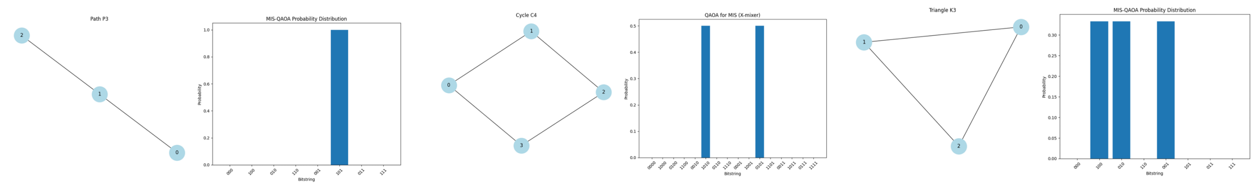
(a) Path P_3 (b) Cycle C_4 (c) Triangle K_3

Figure 2: Computational basis state probability distributions for Maximum Independent Set (MIS) under penalty-based QAOA at depth $p = 2$. (a) For P_3, the state converges deterministically to the ground-state solution $|101\rangle$ with probability 1.00. (b) For C_4, the distribution captures the Z_2 symmetry of the graph, yielding equal peak probabilities of 0.43 for the degenerate optimal bitstrings $|0101\rangle$ and $|1010\rangle$. (c) For K_3, geometric frustration leads to a symmetric three-way split (≈ 0.33 each) across the ground-state cuts $|100\rangle$, $|010\rangle$, and $|001\rangle$.

Table 5: Penalty-based QAOA for MIS ($\lambda = 2$): approximation ratio at $p = 1, 2$ and feasibility rate at $p = 1$.

Graph	MIS size	r ($p = 1$)	r ($p = 2$)	Feasibility rate ($p = 1$)
P_3	2	1.00	1.00	0.72
C_4	2	0.50	0.75	0.68
K_3	1	1.00	1.00	0.55

Table 6 reports the constraint-preserving mixer, implemented as a neighbor-conditioned bit-flip restricted to the independent-set subspace (Section 3.4). All three instances are solved exactly already at $p = 1$, and the feasibility rate is 1.0 throughout by construction. Table 7 compares both strategies directly on C_4: the constraint-preserving mixer achieves a strictly higher approximation ratio and perfect feasibility without penalty tuning, at roughly double the CNOT count of the penalty method.



(a) Path P_3 (b) Cycle C_4 (c) Triangle K_3

Figure 3: Computational basis state probability distributions for Maximum Independent Set (MIS) under the modified mixer-updating framework. (a) For P_3, the state collapses deterministically to the unique ground state $|101\rangle$ ($p = 1.00$). (b) For C_4 at circuit depth $p = 2$, the distribution splits equally (0.50 each) between the degenerate optimal configurations $|1010\rangle$ and $|0101\rangle$. (c) For K_3, structural frustration yields a symmetric three-way split (≈ 0.33 each) across the single-vertex independent sets $|100\rangle$, $|010\rangle$, and $|001\rangle$, with invalid states strictly suppressed.

Table 6: QAOA with the constraint-preserving mixer for MIS: approximation ratio at $p = 1, 2$ and feasibility rate.

Graph	MIS size	r ($p = 1$)	r ($p = 2$)	Feasibility rate
P_3	2	1.00	1.00	1.00
C_4	2	1.00	1.00	1.00
K_3	1	1.00	1.00	1.00

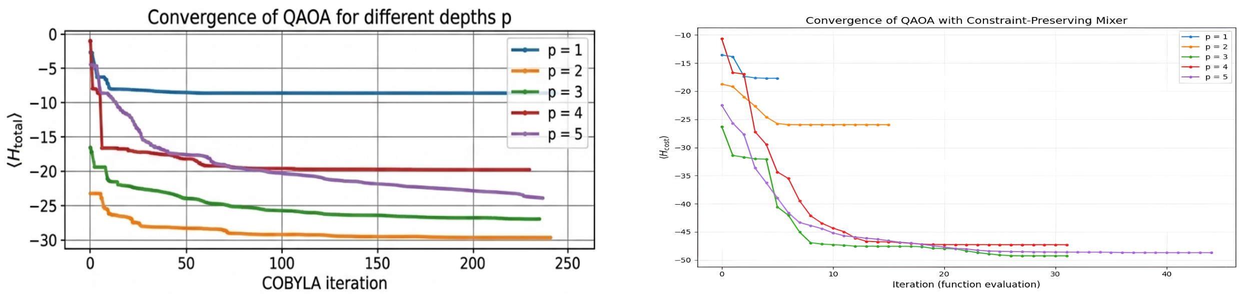
Table 7: Penalty method versus constraint-preserving mixer for MIS on C_4.

Metric	Penalty ($p = 2$)	Constraint-preserving ($p = 1$)
Approximation ratio	0.75	1.00
Feasibility rate	0.68	1.00
Circuit depth (CNOT count)	~ 12	~ 24
Requires penalty tuning	Yes	No

Table 8 evaluates the constraint-aware, penalty-based RQAOA of Section 4.4 against standard penalty QAOA at $p = 2$. On P_4, both methods reach the optimum. On C_5, standard QAOA plateaus at $r = 0.75$, while RQAOA using only $p = 1$ inner circuits and the modified elimination rule of Eq. (16) recovers the exact MIS ($r = 1.00$) with a high feasibility rate (0.9), since every elimination step is independence-preserving by construction. This is the central qualitative result for MIS: recursive, correlation-guided elimination recovers instances that a fixed, shallow ansatz cannot resolve, without requiring deeper circuits.

Table 8: Constraint-aware penalty-based RQAOA (inner $p = 1$) versus standard penalty QAOA ($p = 2$) for MIS.

Graph	MIS size	QAOA ($p = 2$), r	RQAOA (inner $p = 1$), r	Feasibility (RQAOA)
P_4	2	1.00	1.00	1.00
C_5	2	0.75	1.00	0.90



(a) Penalty-based QAOA

(b) Constraint-preserving mixer QAOA

Figure 4: Optimization trajectory and convergence performance as a function of classical optimizer iterations across circuit depths $p \in \{1, 2, 3, 4, 5\}$. (a) Convergence of the standard penalty-based QAOA framework using COBYLA optimization. (b) Rapid, monotonic convergence under the constraint-preserving mixer framework, demonstrating significantly faster ground-state energy minimization across evaluated circuit depths.

6.2 Balanced Weighted MaxCut

Figure 4 summarizes convergence and approximation-ratio scaling with circuit depth for penalty QAOA and the constraint-preserving XY-mixer on the traffic grid of Fig. 1. Penalty QAOA improves monotonically but slowly with depth; the constraint-preserving mixer improves from $r \approx 0.66$ at $p = 1$ to a peak near $r \approx 0.95$ at $p = 3$. Figure 5 shows the corresponding result for constraint-aware penalty-based RQAOA: the approximation ratio is $r = 1.0000$ at every tested depth, including $p = 1$, indicating that the weight-merging elimination rule of Eq. (17) recovers the exact balanced partition without requiring any increase in per-iteration circuit depth.

Table 9 reports the matched-instance headline comparison. Penalty-based RQAOA attains the perfect approximation ratio $r = 1.0000$ using only a single QAOA layer, exceeding the constraint-preserving mixer's $r = 0.9478$ at $p = 3$ and standard penalty QAOA's $r = 0.8341$ even at $p = 5$. Figure 6 shows the resulting optimal partition.

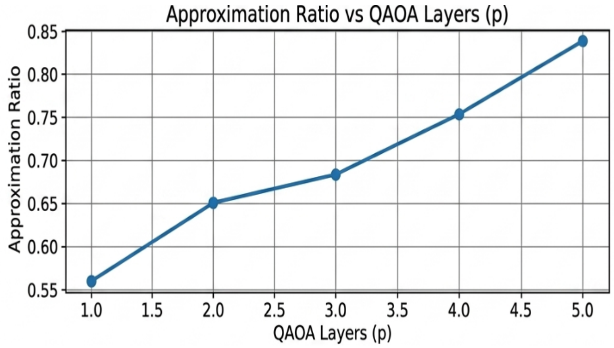
Table 9: Matched-instance comparison of the three QAOA-based methods on the balanced weighted MaxCut traffic-partitioning instance (Fig. 1).

Depth p	Algorithmic approach	Cut value	Approximation ratio
1	Constraint-aware penalty RQAOA	55.00	**1.0000**
3	Constraint-preserving XY-mixer QAOA	52.13	0.9478
5	Standard penalty QAOA	45.88	0.8341

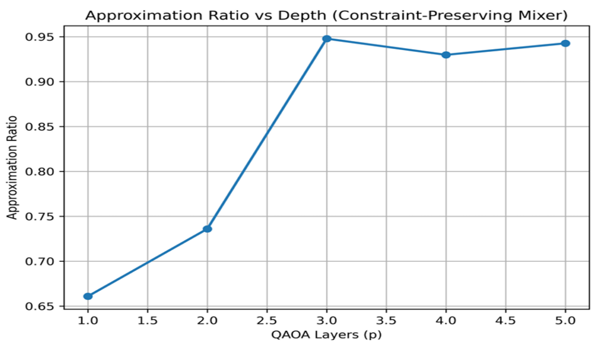
6.3 Comparative Analysis

Three consistent patterns emerge across the unconstrained MaxCut baseline (Tables 2,4) and the two constrained applications (Sections 6.1,6.2).

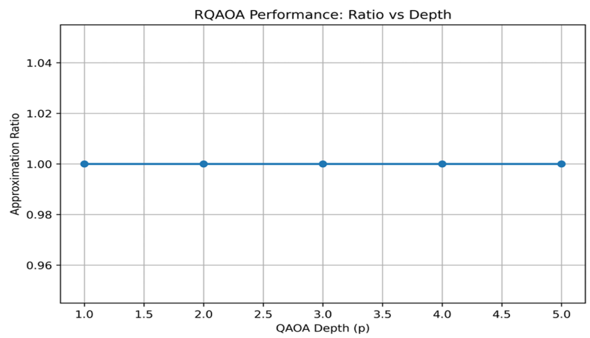
Recursion substitutes for depth. In every problem setting tested, RQAOA with inner depth $p = 1$ matches or exceeds the approximation ratio of fixed-ansatz QAOA at substantially higher depth: $r = 1.0000$ (RQAOA, $p = 1$) versus $r = 0.9347$ (QAOA, $p = 5$) on the five-vertex MaxCut instance; $r = 1.00$ (RQAOA, inner $p = 1$) versus $r = 0.75$ (QAOA, $p = 2$) on C_5-MIS;



(a) Penalty-based QAOA



(b) Constraint-preserving mixer QAOA



(c) Recursive QAOA (RQAOA)

Figure 5: Approximation ratio performance across increasing circuit depths $p \in \{1, 2, 3, 4, 5\}$. (a) Standard penalty-based QAOA showing steady scaling with circuit depth. (b) Constraint-preserving mixer QAOA achieving high approximation ratios at lower depths ($p \geq 3$). (c) Recursive QAOA (RQAOA) maintaining a optimal approximation ratio of 1.00 independently of circuit depth due to variable elimination.

and $r = 1.0000$ (RQAOA, $p = 1$) versus $r = 0.8341$ (penalty QAOA, $p = 5$) and $r = 0.9478$ (constraint-preserving mixer, $p = 3$) on the traffic-partitioning instance. This is consistent with the light-cone picture of Proposition 1: increasing p extends the effective causal neighborhood of a fixed ansatz, whereas recursive elimination propagates long-range structure directly, without deepening any individual circuit.

Constraint-aware elimination is necessary, not incidental. The unconstrained RQAOA elimination rule of Algorithm 2 is not, by itself, feasibility-preserving once a hard or aggregate constraint is introduced; the modified rules of Section 4.4 restricting MIS elimination to negative correlations, and merging edge weights under vertex removal for balanced MaxCut are what allow RQAOA's depth advantage to transfer to the constrained setting without sacrificing feasibility. This is reflected in the feasibility rates of Table 8 (0.901.00 for RQAOA versus 0.550.72 for unmodified penalty QAOA in Table 5).

The trade-off is structure-dependent, not universal. RQAOA's advantage is largest on graphs with moderate size and non-trivial structure (cycles, grids); on highly symmetric graphs such as K_4, RQAOA offers little beyond standard QAOA at $p = 2$ (Table 3), because all pairwise correlations are equal and the elimination order becomes immaterial. Similarly, the constraint-preserving mixer strictly dominates the penalty method in solution quality (Table 7) but at roughly double the CNOT count, so the preferred strategy in practice depends on whether a given NISQ device is more

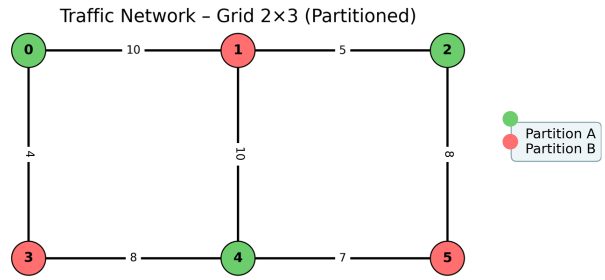


Figure 6: Optimal balanced partition of the traffic network: Partition A (vertices $0, 2, 4$) versus Partition B (vertices $1, 3, 5$). All three methods can in principle reach this configuration given sufficient depth or optimization budget, but constraint-aware RQAOA reaches it with the least circuit resource, requiring only $p = 1$ (Table 9).

depth-limited or gate-error-limited. Constraint-aware RQAOA, evaluated here, occupies a favorable point in this trade-off space: it matches or exceeds the feasibility of the constraint-preserving mixer while using only the shallow, standard-mixer circuits of the penalty method.

7 Limitations

The scope of this study is bounded in several respects that should inform how the results above are read.

- **Noiseless simulation only.** All results are obtained from Qiskit's noiseless statevector and QASM simulators; no gate errors, decoherence, or readout noise are modeled. The comparative advantage of constraint-aware RQAOA over fixed-ansatz methods established here in terms of circuit depth has direct relevance to NISQ hardware, where depth correlates with accumulated error, but the magnitude of that advantage on physical devices remains to be confirmed experimentally.

- **Small problem sizes.** Instances are restricted to $n \leq 8$ vertices, the practical limit for exact classical statevector simulation used for both correctness verification and parameter optimization. Scaling behavior of the closed-form analysis (Proposition 1) and of the recursive framework (Algorithm 3) to graphs of practical, industrial size has not been demonstrated.

- **Heuristic penalty tuning.** The penalty coefficients λ (MIS) and λ_B (balanced MaxCut) are tuned by hand rather than derived analytically, consistent with a broader open problem in penalty-based QUBO encodings [1].

- **Scope of the analytical result.** Proposition 1 is established for $p = 1$ and the unweighted MaxCut cost Hamiltonian; it is used here for numerical validation and as structural motivation for correlation-based elimination, but is not itself extended analytically to the penalty Hamiltonians of Section 4.3 or to $p \geq 2$.

- **Structure-dependent advantage.** As discussed in Section 6.3, RQAOA (constrained or unconstrained) offers little benefit over standard QAOA on highly symmetric or dense graphs, where pairwise correlations are close to uniform and the elimination order becomes immaterial; characterizing this boundary more precisely is left to future work.

8 Conclusion

We studied constrained combinatorial optimization with QAOA and its recursive variant along two complementary axes. Analytically, we derived and numerically validated a closed-form expression for the exact $p = 1$ MaxCut cost expectation on arbitrary graphs, showing that single-layer QAOA treats each edge independently as a function of local degree and shared-triangle count the structural reason fixed, shallow QAOA succeeds on bipartite instances but degrades on graphs whose relevant correlations extend beyond the immediate neighborhood of an edge. Algorithmically, we used this picture to motivate and construct a constraint-aware, penalty-based Recursive QAOA framework, in which the elimination and back-substitution rules that ordinarily assign variables by correlation sign alone are made explicitly feasibility-preserving: a modified rule for the hard independence constraint of Maximum Independent Set, and a weight-merging rule for the soft cardinality-balance constraint of weighted MaxCut partitioning.

Across three problem settings unconstrained MaxCut, MIS, and balanced weighted MaxCut the same pattern holds: recursive, correlation-guided elimination with shallow ($p = 1$) inner circuits matches or exceeds the solution quality of fixed-ansatz QAOA run at substantially greater depth. The traffic-network partitioning case study makes this concrete: constraint-aware RQAOA reached a perfect approximation ratio of 1.0000 using a single QAOA layer, a result that standard penalty QAOA did not reach even at five times the depth ($r = 0.8341$ at $p = 5$), and that a constraint-preserving XY-mixer approached only at three times the depth ($r = 0.9478$ at $p = 3$). Taken together with the systematic comparison of penalty-based and constraint-preserving strategies for MIS (Section 6.1), these results support a general conclusion: for constrained combinatorial optimization on near-term devices, algorithmic structure recursive elimination guided by measured correlations can substitute for the circuit depth that fixed-ansatz methods would otherwise require, provided the elimination rule itself is designed to respect the problem's constraints.

These findings also delimit clear next steps, several of which follow directly from the limitations of Section 7: extending the closed-form analysis beyond $p = 1$ and to penalized (constrained) Hamiltonians directly; replacing the fixed correlation threshold with an adaptive or magnetization-informed elimination rule; benchmarking the constraint-aware framework under realistic noise models and, ultimately, on physical NISQ hardware against classical baselines such as GoemansWilliamson or simulated annealing at matched problem size; and applying the same constraint-encoding methodology penalty Hamiltonian plus feasibility-preserving recursive elimination to other constrained combinatorial problems of practical interest, including resource-constrained scheduling and vehicle routing, and to city-scale traffic networks built from empirical rather than synthetic traffic data.

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Data Availability

The source code, simulation data, and supplementary materials supporting the findings of this study are publicly available in the GitHub repository: https://github.com/Sain123-husfy/QAOA_

Constraint

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